

# MELİH ÜÇER

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<https://melih-ucer.github.io>

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| EDUCATION              | <b>Bilkent University</b><br>PhD in Mathematics<br><i>Thesis:</i> Alexander Modules of Trigonal Curves<br><i>Advisor:</i> Alex Degtyarev                                                                                                                             | 02/2021           |
|                        | <b>Massachusetts Institute of Technology</b><br>Bachelor of Science in Physics                                                                                                                                                                                       | 06/2014           |
| EMPLOYMENT             | <b>KAUST (King Abdullah University of Science and Technology)</b><br>Postdoctoral Researcher in the Mean Field Games Group<br><i>Supervisor:</i> Diogo Gomes                                                                                                         | 07/2023 – present |
|                        | <b>Ankara Yıldırım Beyazıt University</b><br>Taught Undergraduate Engineering Mathematics                                                                                                                                                                            | 01/2015 – 07/2023 |
| RESEARCH INTERESTS     | Mean Field Games, Analysis of Nonlinear PDE, Algebraic Topology                                                                                                                                                                                                      |                   |
| PAPERS                 | <b>Existence and Structure for First-Order Time-Dependent Mean-Field Games with Local Couplings</b><br>Preprint, awarded Best Paper in Dynamic Games:<br><a href="http://arxiv.org/abs/2606.28378">http://arxiv.org/abs/2606.28378</a><br><i>Coauthors:</i> D. Gomes |                   |
|                        | <b>Ranking Mean Field Planning Games</b><br>Accepted for Nonlinear Analysis:<br><a href="https://arxiv.org/abs/2603.02921">https://arxiv.org/abs/2603.02921</a><br><i>Coauthors:</i> A. Almadeh, T. Bakaryan, D. Gomes                                               |                   |
|                        | <b>Solving Mean-Field Games with Monotonicity Methods in Banach Spaces</b><br>Accepted for SIAM Journal of Mathematical Analysis:<br><a href="https://arxiv.org/abs/2506.21212">https://arxiv.org/abs/2506.21212</a><br><i>Coauthors:</i> R. Ferreira, D. Gomes      |                   |
|                        | <b>Regularity for Weak Solutions to First-Order Local Mean-Field Games</b><br>Nonlinear Differ. Equ. Appl. <b>32</b> , 99 (2025)<br><i>Coauthors:</i> A. Alharbi, G. Di Fazio, D. Gomes                                                                              |                   |
|                        | <b>A Mean-Field Game Model of Price Formation With Price-Dependent Agent Behavior</b><br>Sao Paulo J. Math. Sci. <b>18</b> , 1081-1094 (2024).<br><i>Coauthors:</i> K. Aljadhai, M. Almarhoumi, D. Gomes                                                             |                   |
|                        | <b>On the Alexander invariants of trigonal curves</b><br>Rev. Mat. Complut. <b>35</b> , 265–286 (2022).                                                                                                                                                              |                   |
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| SELECTED INVITED TALKS | <b>21st International Symposium on Dynamic Games and Applications</b>                                                                                                                                                                                                | 07/2026           |
|                        | <b>15th AIMS Conference - Special Session on Mean Field Games</b>                                                                                                                                                                                                    | 07/2026           |
|                        | <b>Bolzano Workshop on Mean Field Models and Optimal Control</b>                                                                                                                                                                                                     | 06/2026           |

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|                            | <b>KAUST Workshop on Analysis and PDEs</b>                                                                                                                                                                                                                                                                                                        | 04/2025              |
|                            | <b>Durham Symposium on Mean-Field Games</b>                                                                                                                                                                                                                                                                                                       | 03/2025              |
|                            | <b>14th AIMS Conference - Special Session on Mean Field Games</b>                                                                                                                                                                                                                                                                                 | 12/2024              |
|                            | <b>PDMI Rohlin Topology Seminar (online)</b>                                                                                                                                                                                                                                                                                                      | 11/2020              |
|                            | <b>Ankara-Istanbul Algebraic Geometry &amp; Number Theory Meetings</b><br>A series of three talks.                                                                                                                                                                                                                                                | Fall 2019            |
| <b>SELECTED<br/>HONORS</b> | <b>Başar-Olsder Award for Best Paper in Dynamic Games</b><br>Awarded by International Society of Dynamic Games                                                                                                                                                                                                                                    | 2026                 |
|                            | <b>Alisbah Best Graduate Student Award</b><br>Awarded by Bilkent University Mathematics Department.                                                                                                                                                                                                                                               | 2020                 |
|                            | <b>Nationwide 1st rank</b><br>ALES: Turkish Graduate Schools Entrance Exam                                                                                                                                                                                                                                                                        | Fall 2014            |
|                            | <b>Mathematical Olympiad Medals</b><br>Multiple gold, silver, and bronze medals in international, regional, and national olympiads.                                                                                                                                                                                                               | 2004 –<br>2010       |
| <b>OUTREACH</b>            | <b>Mathematical Olympiad Training</b><br>Instructor at national training camps (Turkey, Saudi Arabia) and occasional instructor in several other countries (Azerbaijan, Palestine, Iran, UAE). Author of problems for Turkish National Olympiads. Served in Turkish delegation at international olympiads as leader, deputy leader, and observer. | 02/2015 –<br>present |
|                            | <b>Problem Development for National Examinations (Turkey)</b><br>Authored mathematics problems for standardized tests including the National University Entrance Exam.                                                                                                                                                                            | 01/2018 –<br>07/2023 |